NUSGRIP

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All questions must be answered, unless marked optional.

Details

Innovator

CV

Idea Bank

Objective

Additional Attachments

Idea Bank

Idea Bank is a platform providing participants the opportunity to solve industry problems and bringing available technology to market.

Deep Technology for Hire!!!

A ready list of technology on http://technology.nus.edu.sg/ is waiting to move out of the lab into the market. We are seeking energetic innovators to embark on this exciting journey.

To learn more about the individual technology, contact the technology manager listed on http://technology.nus.edu.sg/

Technology Highlights

Inventors of the following technology are looking for passionate individuals to drive its commercialization:

Please stay tuned.

I am keen to explore existing technologies.

Yes

I am interested to work on Technology #

2018-004

E.g.. Technology #2017-342, Technology #2018-215

Please refer to http://technology.nus.edu.sg/ for the list of technologies.

For Robotic Tiling Machine, please enter R002-TH001.

Technology Title

Novel Power Management Chip Module to Enable Operation without Battery Energy

The Technology Title should be linked to Technology # in the Technology Database or Technology Highlights.

E.g.."Wireless e-textiles" is the title for technology # 2018-237 and
"TMDs Quantum dots" is the title for technology # 2018-215 as seen at http://technology.nus.edu.sg/ .

Please refer to http://technology.nus.edu.sg/ for the list of technologies.

For Robotic Tiling Machine, please enter "Robotic Tiling Machine".

Technology Abstract 76 / 100 words

B I H | About Markdown formatting

NUS' batteryless module for integrated circuits enables the uncommon capability to uninterruptedly sense, process, capture and timestamp events of interest and for such valuable data to be wirelessly transmitted to the cloud when battery becomes available again. Despite being

in minimum-power mode when battery is not available, the reduced speed of microchip incorporating NUS' technology is still adequate for numerous IoT applications that sense parameters varying slowly in time such as temperature, humidity, light, and pressure.

You can copy the technology abstract you are interested in from the relevant sites. E.g., http://technology.nus.edu.sg/

Why do you want to take this technology further? 59 / 100 words

B I H | About Markdown formatting

Before pursuing my masters degree at NUS, I worked with start-ups that were developing remote renewable energy management systems. Through these experiences, I saw the challanges and short-falls of current IoT solutions. I also saw the opportuntiy to develop new technology that can be applied in not only energy management but also in transportation, buildings, and smart home systems.

E.g., I have strong technical expertise in this field and I can bring it to market; I know an industrial problem with high commercial value that can be solved by this technology.

Describe the NEED. 189 / 350 words

B I H | About Markdown formatting

The customer need will become clearer during the customer discovery process. Nonetheless, I have a hypothesis about what the underlying need is across various customer segments. The technology description page summarizes the challange of IoT devices quite well. The devices need to be able to operate remotely for long-periods of time with minimal access to energy (e.g., periodic battery replacement or charging). This creates a challange for many commercial applictions since regular device charging or battery replacement will add to the overhead costs of managing an IoT network. Combined hardware-software solution can make this problem more managable. Customers can uncover the value of Big Data generated by IoT devices by deploying an IoT system that reduces the maintenance overhead while still providing critical data. This is applicable across industires and various smart citiy initiatives that aim to integrate the physical and digital systems -- this device would be a critical interface between the two. A clear appication area is in the topic of remote energy monitoring, agricultural soil sensing, building temperature regulation, and traffic detection.

The underlying mechanism behind this technology is applicable to a broad range range of industries.

- What is the problem faced today that you want to solve?
- Why do you want to solve this problem?
- What is the business opportunity in solving this problem?

Describe the APPROACH.

188 / 350 words

B I H | About Markdown formatting

Although I have a hypothesis about the and how this technology can solve them, the approach to developing the product will become clearer as I speak with more customers and take a deep dive into the technology itself. In general, I would follow established entreprenurial methods that include market research, customer discovery, product discovery, prototype development, and product validation. After a few iterations, I would ideally have arrived at a compelling product offering that customers are willing to purchase. Then, I would start planning for production and pilot demonstrations with anchor customers. A complex IoT-enabled project may take 6 to 12 months to fully pilot, depending on its scale and complexity. Throughout this process, technology will be fine-tuned based on new findings about and the market. Moreover, I will likely need to bring on additional part-time or full-time team members. If this is executed well, the product will catch an interest from many customers then the focus will be gradually shifted to scaling operations. All in call, a solid approach starts with a good understanding of and a valuable technology-oriented product offering.

How do you intend to solve the problem?

The section below is optional. You are encouraged to attempt them as much as you can. However, if you unable to answer them, it will not affect your application.

Please submit your response to the questions in this section in a separate document if you would like to include Tables, Charts and Figures etc. You can upload your document in the "Attachments" tab.

If you like to attempt them but need some help, please contact us by email at grip@nus.edu.sg.

Describe the quantitative benefit of such Approach to the customer and corresponding cost to the customer in adopting this Approach. (optional)

60 / 300 words

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I would be happy to put together some preliminary market research, but I would need to get a sense of which industry to focus on. This technology can be applied across several sectors. Alternatively, the product offering can be industry-agnostic, meaning the technology will be geared towards a broad range of users. These decisions will take some time to make.

Describe the competing solutions out there. (optional)

39 / 300 words

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IoT applications are a rapidly growing industry and come with many competitors ranging from big players such as AWS and GE to smaller start-ups. The focus on an application area or industry will guide who the immediate competitors are.

If you are familiar with business model canvas, please submit a business model canvas on your project and a narrative on how the 9 pieces fit together. (optional)

17 / 300 words

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I could put this together if the technology is still available for commercialization through the GRIP programme.

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